

Full Stack Development with MERN

Project Documentation (FSD)

Date	15-07-2026
Team ID	SWTID-2026-3345
Project Name	BookVerse – Online Book Store
Documentation Version	v1.0.0 (Release)

1. Introduction

Project Title: BookVerse – Online Book Store

Team Members & Roles:

The BookVerse project is developed by a dedicated MERN stack engineering team, with roles allocated to cover end-to-end full-stack processes:

- Boodi Puneeth Reddy (Team Leader): Full-Stack Architecture, consolidated checkout, separate tracking timelines, database schemas, and stock status updates.
- Parandama Boorsu (Team Member): UI/UX Architecture, checkout panels, progress timelines, and account authorization UI.
- Abhishek Kalluru (Team Member): QA & Backend Developer, low-stock indicators, administrator tables, catalog statistics, and testing.

2. Project Overview

Purpose: BookVerse – Online Book Store is a multi-vendor bookstore platform. It allows buyers to purchase books from multiple sellers under a single checkout session, reducing shipping steps. The system tracks delivery status for each book item separately and displays catalog statistics for administrators and sellers on their respective dashboard panels.

Features: • Single-Checkout Multi-Vendor Consolidation: Merges shopping carts containing Biographies, Fiction, and Science books from different sellers under a unified order.

- Order Tracking: Allows each book in an order to progress through fulfillment states separately (Pending, Confirmed, Packed, Shipped, Delivered).

- **Delivery Estimates:** Calculates delivery dates when the seller accepts an order.
- **Dashboard Statistics:** Shows total count of active listings and total sales revenue on the dashboard.

3. Architecture

Frontend React Architecture: The frontend client is implemented as a Single Page Application (SPA) using React.js built with Vite. It incorporates responsive designs, Axios for API communication, and React Router DOM for route controls.

Backend Express Architecture: The backend API is modeled as an Express.js REST application running on Node.js. It implements the MVC architectural pattern consisting of route controllers, body parsers, file upload handling, and session auth middleware.

Database MongoDB Schema: The database layer is hosted on MongoDB Atlas and integrated using Mongoose. Schemas are modeled with Mongoose validations and relationship references. Stock counts are updated automatically when delivery is marked complete.

4. Setup Instructions

Prerequisites:

- Node.js (v18.0.0 or higher)
- npm (v9.0.0 or higher)
- MongoDB Atlas Cluster URL
- Git command line client

Installation Guide:

Follow these step-by-step setup guides to install the codebase locally:

```
# 1. Clone the repository
git clone https://github.com/BoodiPuneethReddy/BookVerse-SkillsWallet.git
cd BookVerse-SkillsWallet

# 2. Install Server Dependencies
cd server
npm install

# 3. Configure Server Environment Variables (.env)
PORT=5000
MONGO_URI=<MongoDB Atlas Connection String>
JWT_SECRET=<Secure JWT Secret>
CLOUDINARY_API_KEY=<Cloudinary API Key>

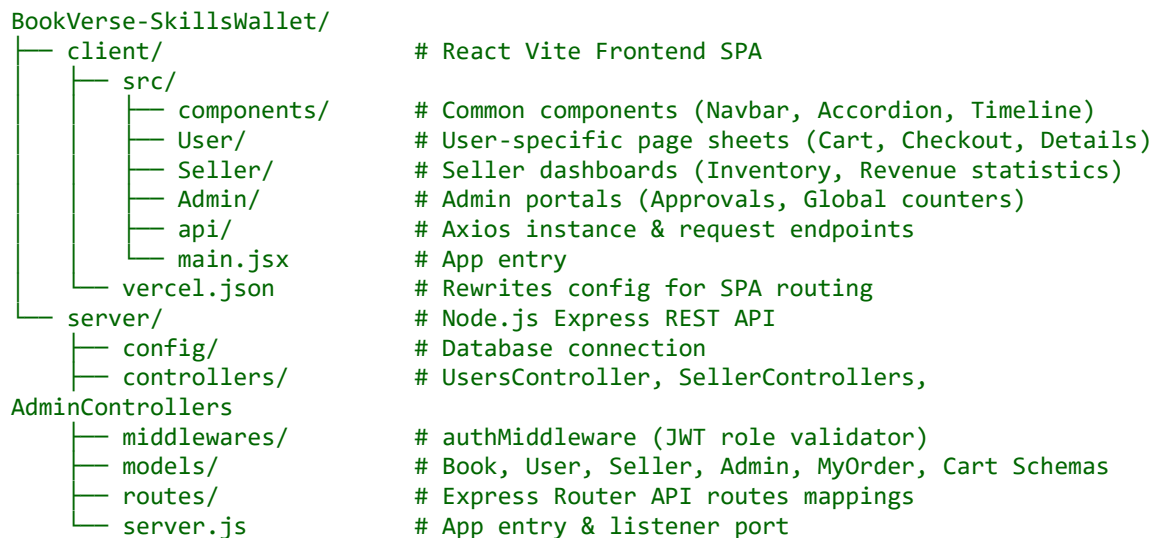
# 4. Install Client Dependencies
cd ../client
npm install

# 5. Configure Client Environment Variables (.env)
VITE_API_URL=http://localhost:5000/api
```

5. Folder Structure

Project Workspace Structure:

The folder structure displays clear modular organization for both React frontend client and Express backend server layers:



6. Running the Application

Execution Commands:

To execute the application locally, run both client and server development servers concurrently:

1. Backend Server:

Execute this command in the server directory to start the Express API backend server:

```
cd server
npm start
```

2. Frontend Client:

Execute this command in the client directory to start the Vite React development server:

```
cd client
npm run dev
```

The web UI will be accessible locally at <http://localhost:5173>, and the Express API server will listen on <http://localhost:5000>.

7. API Documentation

Exposed API Routes:

The API exposes REST endpoints protected by role middlewares where applicable:

• POST /api/users/login (Public)

Authenticates credentials. Returns user role and session token payload.

```
Request Body:
{
```

```

    "email": "user@bookstore.com",
    "password": "userpassword"
  }
  Response (200 OK):
  {
    "success": true,
    "token": "eyJhbGciOi...",
    "role": "User"
  }

```

- **POST /api/orders (Protected - User)**

Consolidates cart items and submits a single unified order.

```

Request Body:
{
  "items": [
    { "book": "65d21a...", "seller": "65d21b...", "quantity": 2 }
  ]
}
Response (201 Created):
{
  "success": true,
  "orderId": "65d22f..."
}

```

- **PUT /api/seller/orders/:id/status (Protected - Seller)**

Updates the status timeline of a specific book item in an order. Triggers stock updates when transitioned to 'Delivered'.

```

Request Body:
{
  "itemId": "65d22a...",
  "status": "Delivered"
}
Response (200 OK):
{
  "success": true,
  "message": "Status updated. Stock decremented successfully."
}

```

8. Authentication

The system uses token-based session management. Upon login, the server generates a token representing the user's role and account. The client stores this session token locally and includes it in the authorization headers of API requests. The backend parses this token to restrict or permit access to seller and administrator dashboard endpoints.

9. User Interface

The user interface displays modern layout aesthetics. We have attached key screenshots showcasing critical workflow segments below:

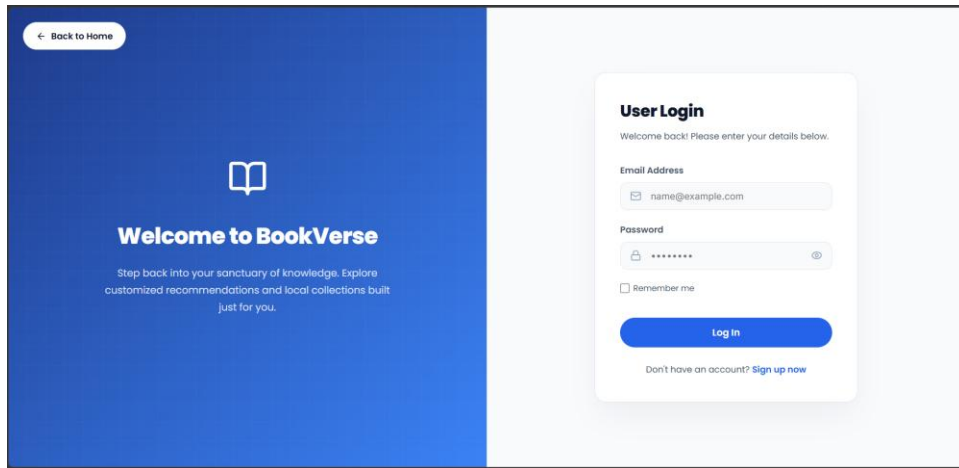


Figure 9.1 – User Login Page

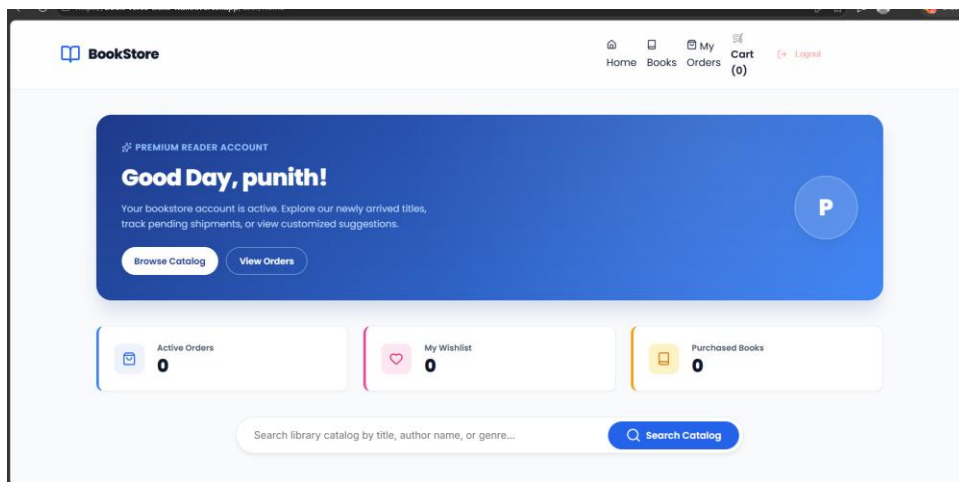


Figure 9.2 – User Dashboard / Home Page

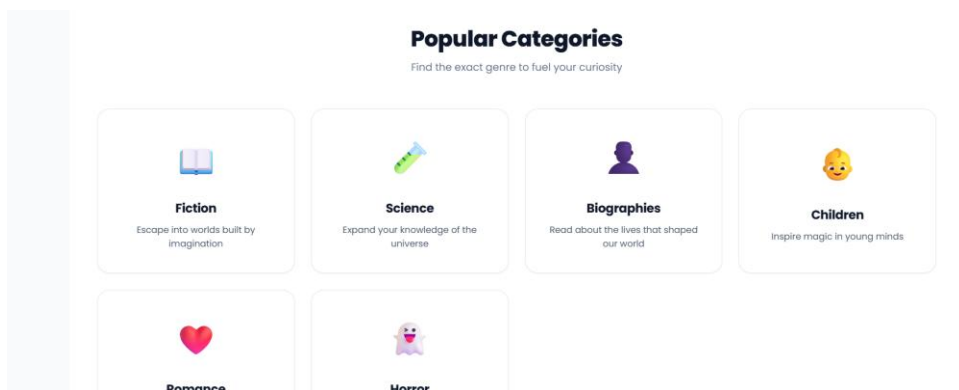


Figure 9.3 – Book Categories & Catalog

Shopping Cart



Atomic Habits
 by James Clear
BookVerse Store

- 1 +

₹450.00



Think and Grow Rich
 by Napoleon Hill
BookVerse Store

- 1 +

₹299.00

Order Summary

Subtotal (2 items)	₹749.00
Shipping Delivery	FREE
Platform Fee	₹15.00
Total Amount	₹764.00

Proceed to Buy Cart →

Figure 9.4 – Shopping Cart

Back

Checkout

Delivery Address

Payment Option

☒ Cash on Delivery (COD)
 ☐ UPI Payment
 ☐ Credit/Debit Card

Review Book Items



Atomic Habits
 by James Clear
 Price: ₹450.00 | Qty: 1
Seller: BookVerse Store



Think and Grow Rich
 by Napoleon Hill
 Price: ₹299.00 | Qty: 1
Seller: BookVerse Store

Order Total Summary

Items Subtotal (2)	₹749.00
Shipping Delivery	FREE
Platform Fee	₹15.00
Grand Total	₹764.00


Estimated Delivery: Waiting for Seller Confirmation
Customer schedules will be updated once each seller confirms stock.

Place Your Order

Figure 9.5 – Checkout & Payment

Your Orders				
ORDER ID	DATE PLACED	TOTAL PRICE	ITEMS COUNT	
#237EE6FE	16/7/2026	₹749.00	2 Books	PENDING
#237EE6FA	16/7/2026	₹749.00	2 Books	PENDING
#ABA1A9A8	15/7/2026	₹749.00	2 Books	CONFIRMED
#ABA1A99D	15/7/2026	₹749.00	2 Books	PACKED

Figure 9.6 – User Orders

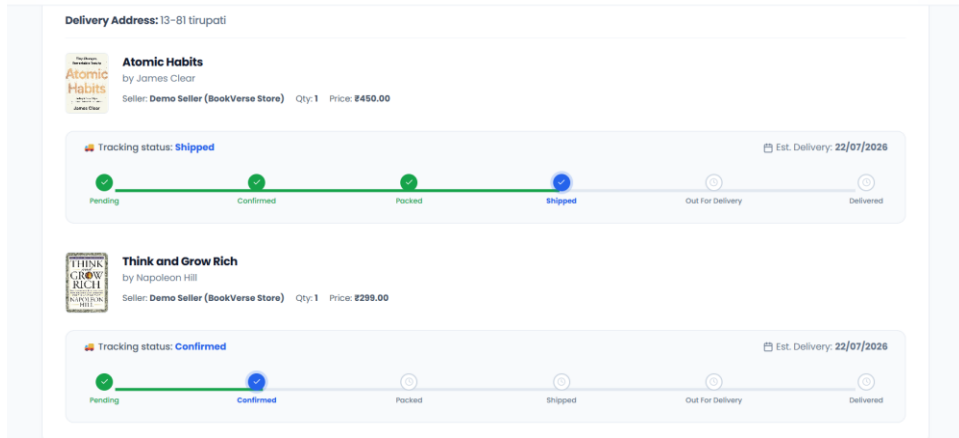


Figure 9.7 – Order Tracking Timeline

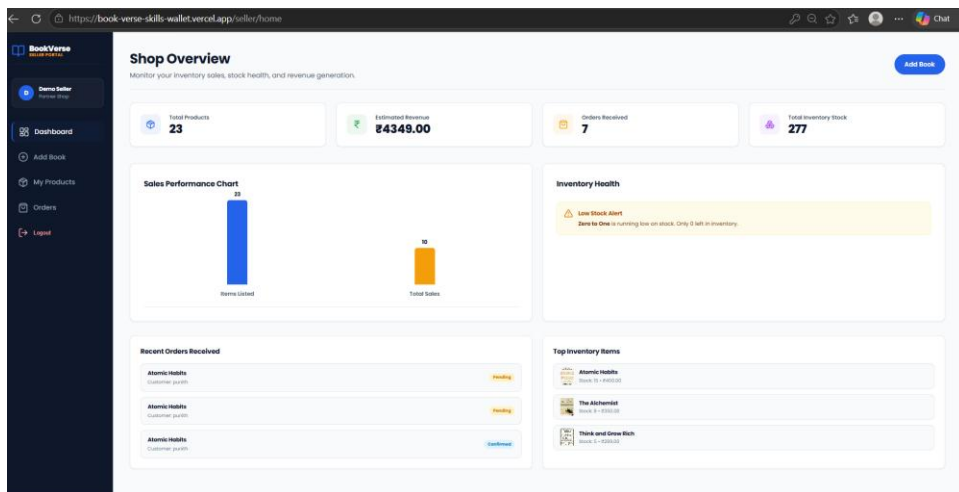


Figure 9.8 – Seller Dashboard

My Products

Book Image	Title	Author	Category	Stock	Price	Actions
	Atomic Habits	James Clear	Biographies	15	₹450.00	View Edit Delete
	The Alchemist	Paulo Coelho	Fiction	8	₹350.00	View Edit Delete
	Think and Grow Rich	Napoleon Hill	Biographies	5	₹299.00	View Edit Delete
	Rich Dad Poor Dad	Robert T. Kiyosaki	Science	20	₹399.00	View Edit Delete
	Rigdi	Hector Garcia	Biographies	12	₹399.00	View Edit Delete
	Deep Work	Cal Newport	Science	10	₹450.00	View Edit Delete
	The Psychology of Money	Morgan Housel	Science	25	₹350.00	View Edit Delete
	The Power of Habit	Charles Duhigg	Science	14	₹400.00	View Edit Delete
	The 7 Habits of highly Effective People	Stephen R. Covey	Biographies	18	₹499.00	View Edit Delete

Figure 9.9 – Seller Product Management

Orders Received								
Product	Order ID	Customer Name	Address	Quantity	Price	Est. Delivery Date	Status	Operation
Atomic Habits	#237EE8FE	punith		1	₹450.00	+ e.g. 24/07/2028	Pending	Update
	#237EE8FE	punith		1	₹299.00	+ e.g. 24/07/2028	Pending	Update
Think and Grow Rich	#237EE8FA	punith		1	₹450.00	+ e.g. 24/07/2028	Pending	Update
	#237EE8FA	punith		1	₹299.00	+ e.g. 24/07/2028	Pending	Update
Atomic Habits	#ABAIA8A8	punith	13-BI trupti	1	₹450.00	24/07/2028	Shipped	Update
Think and Grow Rich	#ABAIA8A8	punith	13-BI trupti	1	₹299.00	24/07/2028	Confirmed	Update
Atomic Habits	#ABAIA8BD	punith	13-BI trupti	1	₹450.00	24/07/2028	Failed	Update

Figure 9.10 – Seller Orders Management

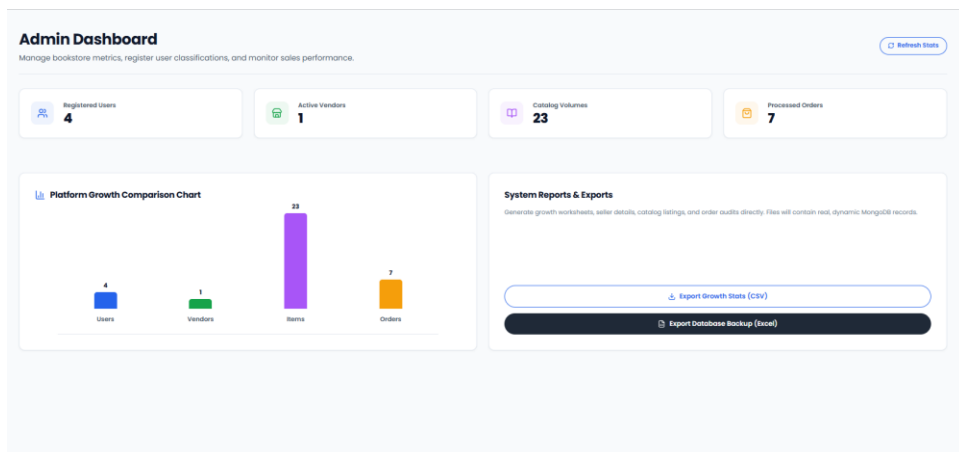


Figure 9.11 – Admin Dashboard

Manage Users					
Sl.No	User ID	Name	Email	Status	Operation
1	6a52828d5eb550f53339ab7	Punith	punith@gmail.com	Active	Block View Orders
2	6a548dab5243356a6e088e	Abhi	abhihihi@gmail.com	Active	Block View Orders
3	6a55ddf2e74fb622009e07		param@gmail.com	Active	Block View Orders
4	6a57ca9517d7f1eab0a0f1b	punith	godofnothing@gmail.com	Active	Block View Orders

Figure 9.12 – Admin User Management

Manage Sellers

Sl.No	Seller	Shop Name	Email	Approval Status	Operation
1	Demo Seller	BookVerse Store	seller@bookverse.com	Approved	Report

Figure 9.13 – Admin Seller Management

Manage Books

Sl.No	Book Cover	Book Title	Author	Category	Seller	Price	Stock	Operation
1		Atomic Habits	James Clear	Biographies	Demo Seller	₹450.00	15	Delete Book
2		The Alchemist	Paulo Coelho	Fiction	Demo Seller	₹350.00	8	Delete Book
3		Think and Grow Rich	Napoleon Hill	Biographies	Demo Seller	₹299.00	5	Delete Book
4		Rich Dad Poor Dad	Robert T. Kiyosaki	Science	Demo Seller	₹399.00	20	Delete Book
5		Higai	Héctor García	Biographies	Demo Seller	₹399.00	12	Delete Book
6		Deep Work	Cal Newport	Science	Demo Seller	₹450.00	10	Delete Book
7		The Psychology of Money	Morgan Housel	Science	Demo Seller	₹350.00	25	Delete Book
8		The Power of Habit	Charles Duhigg	Science	Demo Seller	₹400.00	14	Delete Book
9		The 7 Habits of Highly Effective People	Stephen R. Covey	Biographies	Demo Seller	₹499.00	18	Delete Book

Figure 9.14 – Admin Book Management

10. Testing

The testing phase verified core requirements through User Acceptance Testing (UAT) scripts. Scenarios covered user signup, category filtering (Biographies, Fiction, Science), cart checkouts, order item timelines, delivery date calculations, and stock counts. Results were tracked across Pass, Passed with Observation, and Pending Verification states to ensure reliability before deployment.

11. Screenshots or Demo

The project is fully deployed and accessible at the following live links:

- **Frontend client:** <https://book-verse-skills-wallet.vercel.app>
- **Backend Server REST API:** <https://bookverse-api-64ja.onrender.com>
- **Repository Name:** BookVerse-SkillsWallet
- **Repository URL:** <https://github.com/BoodiPuneethReddy/BookVerse-SkillsWallet>
- **Clone Command:** `git clone https://github.com/BoodiPuneethReddy/BookVerse-SkillsWallet.git`

12. Known Issues

- Password recovery email services are currently mocked on the backend, using local API debug overrides.
- Session invalidation relies entirely on token expiry, missing active server-side blacklist checks.

13. Future Enhancements

- Real-time payment gateway integration (Stripe/Razorpay) during consolidated checkouts.
- Native iOS and Android mobile app wrappers built with React Native sharing the same Express API controllers.
- Book recommendation engine suggesting catalog books based on category purchase histories.